

SOLIDWORKS API (.NET VBA C#) — Reference (202/337): IModelDocExtension (part 8 of 8)

SOLIDWORKS API Documentation (converted from CHM)

IModelDocExtension (part 8 of 8) (Type)

SetKeepLightInRenderScene Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetKeepLightInRenderScene Method (IModelDocExtension)

Sets whether to keep a light when the scene changes.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetKeepLightInRenderScene( _  
    ByVal ID As System.Integer, _  
    ByVal Val As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim ID As System.Integer  
Dim Val As System.Boolean  
  
instance.SetKeepLightInRenderScene(ID, Val)
```

C#

```
void SetKeepLightInRenderScene(  
    System.int ID,  
    System.bool Val  
)
```

C++/CLI

```
void SetKeepLightInRenderScene(  
& System.int ID,  
& System.bool Val  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
Light ID
Val
True to keep a light when the scene changes, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetKeepLightInRenderScene](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetKeepLightInRenderScene Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SetLightEnabledInRender Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetLightEnabledInRender Method (IModelDocExtension)

Sets whether a light is on in this model.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetLightEnabledInRender( _  
    ByVal ID As System.Integer, _  
    ByVal Val As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim ID As System.Integer  
Dim Val As System.Boolean  
  
instance.SetLightEnabledInRender(ID, Val)
```

C#

```
void SetLightEnabledInRender(  
    System.int ID,  
    System.bool Val  
)
```

C++/CLI

```
void SetLightEnabledInRender(  
& System.int ID,  
& System.bool Val  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ID
Light ID
Val
True to enable the light, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetLightEnabledInRender](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetLightEnabledInRender Method](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetMaterialPropertyValues Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetMaterialPropertyValues Method (IModelDocExtension)

Sets the material property values for this model document.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetMaterialPropertyValues( _  
    ByVal Material\_property\_values As System.Object, _
```

```

    ByVal Config\_opt As System.Integer, _
    ByVal Config\_names As System.Object _
)

```

Visual Basic (Usage)



```

Dim instance As IModelDocExtension
Dim Material_property_values As System.Object
Dim Config_opt As System.Integer
Dim Config_names As System.Object

instance.SetMaterialPropertyValues(Material_property_values, Config_opt, Config_names)

```

C#

```

void SetMaterialPropertyValues(
    System.object Material\_property\_values,
    System.int Config\_opt,
    System.object Config\_names
)

```

C++/CLI

```

void SetMaterialPropertyValues(
    & System.Object^ Material\_property\_values,
    & System.int Config\_opt,
    & System.Object^ Config\_names
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Material_property_values

Array of material property values

Config_opt

Configuration option as defined in [swInConfigurationOpts_e](#)

Config_names

Array of configuration names

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetMaterialPropertyValues](#).

Remarks

The material property values include color (R,G,B), reflectivity (ambient, diffuse, specular, shininess), transparency, and emission.

The format of *Material_property_values* is an array of doubles:

[*R*, *G*, *B*, *Ambient*, *Diffuse*, *Specular*, *Shininess*, *Transparency*, *Emission*]

To see a color change, you must:

1. Specify *R*, *G*, and *B*, each with a value between 0.0 and 1.0, inclusive. (These values are internally multiplied by 255.0 to determine the RGB color.)
2. Specify the reflectivity properties (*Diffuse*, *Specular*, *Shininess* (1.0 - Specular spread/Blurriness)), each with a value greater than 0.0 and less than or equal to 1.0.
3. Specify *Ambient*, *Transparency* and *Emission*, each with a value between 0.0 and 1.0, inclusive.
4. Refresh the graphics area after calling this method.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetMaterialPropertyValues Method](#)
[IModelDocExtension::GetMaterialPropertyValues Method](#)
[IModelDocExtension::RemoveMaterialProperty Method](#)
[IModelDocExtension::RemoveMaterialProperty Method](#)
[IModelDocExtension::SetMaterialPropertyValues Method](#)
[IModelDocExtension::HasMaterialPropertyValues Method](#)

Availability

SetSceneBkgDIBx64 Method (IModelDocExtension)

[See Also](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetSceneBkgDIBx64 Method (IModelDocExtension)

Sets the background image in 64-bit applications.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetSceneBkgDIBx64( _
    ByVal L\_dib As System.Long _
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim L_dib As System.Long

instance.SetSceneBkgDIBx64(L_dib)
```

C#

```
void SetSceneBkgDIBx64(
    System.long L\_dib
)
```

C++/CLI

```
void SetSceneBkgDIBx64(
    & System.int64 L\_dib
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

L_dib
Background image as DIBSECTION

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetSceneBkgDIBx64](#).

Remarks

This method is only available through early binding and with 64-bit versions of the SOLIDWORKS Design software.

Old background images are deleted automatically.

For more information about DIBSECTION, see Microsoft's documentation.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetSceneBkgDIBx64 Method](#)
[IModelDoc2::GetSceneBkgDIB Method](#)
[IModelDoc2::SetSceneBkgDIB Method](#)
[IModelDoc2::InsertBkgImage Method](#)
[IModelDoc2::SceneBkgImageFileName Property](#)

Availability

SOLIDWORKS 2006 SP2, Revision Number 14.2

SetSunLightAdvancedPropertyValues Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetSunLightAdvancedPropertyValues Method (IModelDocExtension)

Sets the specified sunlight advanced properties.

Visual Basic Syntax

Visual Basic (Declaration)

```
Function SetSunLightAdvancedPropertyValues( _  
    ByVal Haze As System.Double, _  
    ByVal SunDiameter As System.Double, _  
    ByVal GroundAlbedo As System.Integer, _  
    ByVal SkyGamma As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Haze As System.Double  
Dim SunDiameter As System.Double  
Dim GroundAlbedo As System.Integer  
Dim SkyGamma As System.Double  
Dim value As System.Boolean
```

```
value = instance.SetSunLightAdvancedPropertyValues(Haze, SunDiameter, GroundAlbedo, SkyGamma)
```

C#

```
System.bool SetSunLightAdvancedPropertyValues(  
    System.double Haze,  
    System.double SunDiameter,  
    System.int GroundAlbedo,  
    System.double SkyGamma  
)
```

C++/CLI

```
System.bool SetSunLightAdvancedPropertyValues(  
& System.double Haze,  
& System.double SunDiameter,  
& System.int GroundAlbedo,  
& System.double SkyGamma  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Haze

0.0 <= Haze setting <= 1.0

SunDiameter

0.01 <= sun diameter visible in the scene <= 21474836.47

GroundAlbedo

RGB color reflected from the ground upwards

SkyGamma

0.1 <= visible sky Output Gamma <= 100.0

Return Value

True if successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetSunLightAdvancedPropertyValues](#).

Remarks

After calling this method, call [IModelDocExtension::UpdateSunLight](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetSunLightAdvancedPropertyValues Method \(\)](#)
[IModelDocExtension::SetSunLightSourcePropertyValues Method \(\)](#)
[IModelDocExtension::SunLightInformation Property \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetSunLightSourcePropertyValues Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetSunLightSourcePropertyValues Method (IModelDocExtension)

Sets the property values for a sunlight source.

.NET Syntax

Visual Basic (Declaration)

```
Function SetSunLightSourcePropertyValues( _  
    ByVal NorthDirection As MathVector, _  
    ByVal NorthLatitude As System.Double, _  
    ByVal EastLongitude As System.Double, _  
    ByVal TimeZone As System.Double, _  
    ByVal DateTime As System.String _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim NorthDirection As MathVector  
Dim NorthLatitude As System.Double  
Dim EastLongitude As System.Double  
Dim TimeZone As System.Double  
Dim DateTime As System.String  
Dim value As System.Boolean
```

```
value = instance.SetSunLightSourcePropertyValues(NorthDirection, NorthLatitude, EastLongitude,  
TimeZone, DateTime)
```

C#

```
System.bool SetSunLightSourcePropertyValues(  
    MathVector NorthDirection,  
    System.double NorthLatitude,  
    System.double EastLongitude,  
    System.double TimeZone,  
    System.string DateTime  
)
```

C++/CLI

```
System.bool SetSunLightSourcePropertyValues(  
& MathVector^ NorthDirection,  
& System.double NorthLatitude,  
& System.double EastLongitude,  
& System.double TimeZone,  
& System.String^ DateTime  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

NorthDirection
[IMathVector](#); north direction of the sunlight source
NorthLatitude
North latitude of the sunlight source
EastLongitude
East longitude of the sunlight source
TimeZone
Standard time zone of the sunlight source

DateTime
Date and time stamp in the specified TimeZone

Return Value

True if sunlight source property values are successfully set, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetSunLightSourcePropertyValues](#).

Example

[Get and Set Sunlight Source Property Values \(VBA\)](#)
[Get and Set Sunlight Source Property Values \(VB.NET\)](#)
[Get and Set Sunlight Source Property Values \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::SetLightSourcePropertyValuesVB Method \(\)](#)
[IModelDocExtension::GetSunLightSourcePropertyValues Method \(\)](#)
[IModelDoc2::LightSourcePropertyValues Property \(\)](#)
[IModelDocExtension::SetSunLightAdvancedPropertyValues Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetTexture Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetTexture Method (IModelDocExtension)

Applies texture to the specified configuration.

.NET Syntax

Visual Basic (Declaration)

```
Function SetTexture( _  
    ByVal Config\_name As System.String, _  
    ByVal TextureIn As Texture _  
) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim Config_name As System.String  
Dim TextureIn As Texture  
Dim value As System.Boolean  
  
value = instance.SetTexture(Config_name, TextureIn)
```

C#

```
System.bool SetTexture(  
    System.string Config\_name,  
    Texture TextureIn  
)
```

C++/CLI

```
System.bool SetTexture(  
& System.String^ Config\_name,  
& Texture^ TextureIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Config_name
Name of configuration
TextureIn
[Texture](#)

Return Value

True if texture is applied, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetTexture](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::CreateTexture Method](#)
[IModelDocExtension::GetTexture Method](#)
[IModelDocExtension::RemoveTexture2 Method](#)
[IModelDocExtension::RemoveTextureByDisplayState Method](#)
[IModelDocExtension::SetTextureByDisplayState Method](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

SetTextureByDisplayState Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetTextureByDisplayState Method (IModelDocExtension)

Sets the texture applied to this model in the specified display state.

.NET Syntax

Visual Basic (Declaration)

```
Function SetTextureByDisplayState( _  
    ByVal DisplayStateName As System.String, _  
    ByVal TextureIn As Texture _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim DisplayStateName As System.String  
Dim TextureIn As Texture  
Dim value As System.Boolean  
  
value = instance.SetTextureByDisplayState(DisplayStateName, TextureIn)
```

C#

```
System.bool SetTextureByDisplayState(  
    System.string DisplayStateName,  
    Texture TextureIn  
)
```

C++/CLI

```
System.bool SetTextureByDisplayState(  
& System.String^ DisplayStateName,  
& Texture^ TextureIn  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

DisplayStateName

Name of display state
TextureIn
[Texture](#)

Return Value

True if texture is set on the model, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetTextureByDisplayState](#).

Example

[Apply and Remove Texture To and From Model by Display State \(C#\)](#)
[Apply and Removed Texture To and From Model By Display State \(VB.NET\)](#)
[Apply and Remove Texture To and From Model By Display State \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::SetTexture Method](#)
[IModelDocExtension::RemoveTextureByDisplayState Method](#)
[IModelDocExtension::RemoveTexture2 Method](#)
[IModelDocExtension::GetTexture Method](#)
[IModelDocExtension::CreateTexture Method](#)

Availability

SOLIDWORKS 2010 SP1, Revision Number 18.1

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetTopLevelTransparency Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetTopLevelTransparency Method (IModelDocExtension)

Sets the transparency of this part or top-level assembly.

.NET Syntax

Visual Basic (Declaration)

```
Function SetTopLevelTransparency( _  
    ByVal InValue As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim InValue As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SetTopLevelTransparency(InValue)
```

C#

```
System.bool SetTopLevelTransparency(  
    System.bool InValue  
)
```

C++/CLI

```
System.bool SetTopLevelTransparency(  
    & System.bool InValue  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

InValue

True to make this part or top-level assembly transparent, false to not

Return Value

True if transparency of this part or top-level assembly successfully set, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetTopLevelTransparency](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2018 FCS, Revision Number 26.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetUserPreferenceDouble Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetUserPreferenceDouble Method (IModelDocExtension)

Sets document default user preference values. This method is intended for user preferences of type double.

.NET Syntax

Visual Basic (Declaration)

```
Function SetUserPreferenceDouble( _  
    ByVal UserPref As System.Integer, _  
    ByVal UserPrefOption As System.Integer, _  
    ByVal Value As System.Double _  
    ) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim UserPref As System.Integer  
Dim UserPrefOption As System.Integer  
Dim Value As System.Double  
Dim value As System.Boolean  
  
value = instance.SetUserPreferenceDouble(UserPref, UserPrefOption, Value)
```

C#

```
System.bool SetUserPreferenceDouble(  
    System.int UserPref,  
    System.int UserPrefOption,  
    System.double Value  
)
```

C++/CLI

```
System.bool SetUserPreferenceDouble(  
& System.int UserPref,  
& System.int UserPrefOption,  
& System.double Value  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPref

User preference as defined in [swUserPreferenceDoubleValue_e](#)

UserPrefOption

User-preference option for customizing annotation and dimension drafting standards and as defined in [swUserPreferenceOption_e](#); if not needed, then specify swDetailingNoOptionSpecified

Value

Value to assign to the user preference specified in UserPref

Return Value

True if the user preference is set, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetUserPreferenceDouble](#).

Example

[Get and Set User Preferences \(VBA\)](#)
[Get and Set User Preferences \(VB.NET\)](#)
[Get and Set User Preferences \(C#\)](#)

Remarks

This method is equivalent to interactively setting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::SetUserPreferenceInteger Method](#)
[IModelDocExtension::SetUserPreferenceString Method](#)
[IModelDocExtension::SetUserPreferenceTextFormat Method](#)
[IModelDocExtension::SetUserPreferenceToggle Method](#)
[IModelDocExtension::GetUserPreferenceDouble Method](#)
[IModelDocExtension::GetUserPreferenceDoubleValueRange Method](#)
[IModelDocExtension::GetUserPreferenceInteger Method](#)
[IModelDocExtension::GetUserPreferenceString Method](#)
[IModelDocExtension::GetUserPreferenceTextFormat Method](#)
[IModelDocExtension::GetUserPreferenceToggle Method](#)
[ISldWorks::GetUserPreferenceDoubleValue Method](#)
[ISldWorks::GetUserPreferenceIntegerValue Method](#)
[ISldWorks::GetUserPreferenceStringListValue Method](#)
[ISldWorks::GetUserPreferenceStringValue Method](#)
[ISldWorks::GetUserPreferenceToggle Method](#)
[ISldWorks::SetUserPreferenceDoubleValue Method](#)
[ISldWorks::SetUserPreferenceIntegerValue Method](#)
[ISldWorks::SetUserPreferenceStringListValue Method](#)
[ISldWorks::SetUserPreferenceStringValue Method](#)
[ISldWorks::SetUserPreferenceToggle Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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SetUserPreferenceInteger Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetUserPreferenceInteger Method (IModelDocExtension)

Sets document default user preference values. This method is intended for user preferences of type integer.

.NET Syntax

Visual Basic (Declaration)

```
Function SetUserPreferenceInteger( _  
    ByVal UserPref As System.Integer, _  
    ByVal UserPrefOption As System.Integer, _  
    ByVal Value As System.Integer _  
    ) As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim UserPref As System.Integer  
Dim UserPrefOption As System.Integer  
Dim Value As System.Integer
```



```
Dim value As System.Boolean
```

```
value = instance.SetUserPreferenceInteger(UserPref, UserPrefOption, Value)
```

C#

```
System.bool SetUserPreferenceInteger(  
    System.int UserPref,  
    System.int UserPrefOption,  
    System.int Value  
)
```

C++/CLI

```
System.bool SetUserPreferenceInteger(  
& System.int UserPref,  
& System.int UserPrefOption,  
& System.int Value  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPref

User preference as defined in [swUserPreferenceIntegerValue_e](#)
UserPrefOption

User-preference option for customizing annotation and dimension drafting standards and as defined in [swUserPreferenceOption_e](#); if not needed, then specify swDetailingNoOptionSpecified

Value

Value to assign to the user preference specified in UserPref

Return Value

True if the user preference is set, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetUserPreferenceInteger](#).

Example

[Get and Set User Preferences \(VBA\)](#)
[Get and Set User Preferences \(VB.NET\)](#)
[Get and Set User Preferences \(C#\)](#)

Remarks

This method is equivalent to interactively setting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::SetUserPreferenceDouble Method](#)
[IModelDocExtension::SetUserPreferenceString Method](#)
[IModelDocExtension::SetUserPreferenceTextFormat Method](#)
[IModelDocExtension::SetUserPreferenceToggle Method](#)
[IModelDocExtension::GetUserPreferenceDouble Method](#)
[IModelDocExtension::GetUserPreferenceDoubleValueRange Method](#)
[IModelDocExtension::GetUserPreferenceInteger Method](#)
[IModelDocExtension::GetUserPreferenceString Method](#)
[IModelDocExtension::GetUserPreferenceTextFormat Method](#)
[IModelDocExtension::GetUserPreferenceToggle Method](#)
[ISldWorks::GetUserPreferenceDoubleValue Method](#)
[ISldWorks::GetUserPreferenceIntegerValue Method](#)
[ISldWorks::GetUserPreferenceStringListValue Method](#)
[ISldWorks::GetUserPreferenceStringValue Method](#)
[ISldWorks::GetUserPreferenceToggle Method](#)
[ISldWorks::SetUserPreferenceDoubleValue Method](#)
[ISldWorks::SetUserPreferenceIntegerValue Method](#)
[ISldWorks::SetUserPreferenceStringListValue Method](#)
[ISldWorks::SetUserPreferenceStringValue Method](#)
[ISldWorks::SetUserPreferenceToggle Method](#)

Availability

SetUserPreferenceString Method (IModelDocExtension)

[See Also](#) [Example](#)[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetUserPreferenceString Method (IModelDocExtension)

Sets document default user preference values. This method is intended for user preferences of type string.

.NET Syntax

Visual Basic (Declaration)

```
Function SetUserPreferenceString( _
    ByVal UserPref As System.Integer, _
    ByVal UserPrefOption As System.Integer, _
    ByVal Value As System.String _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim UserPref As System.Integer
Dim UserPrefOption As System.Integer
Dim Value As System.String
Dim value As System.Boolean

value = instance.SetUserPreferenceString(UserPref, UserPrefOption, Value)
```

C#

```
System.bool SetUserPreferenceString(
    System.int UserPref,
    System.int UserPrefOption,
    System.string Value
)
```

C++/CLI

```
System.bool SetUserPreferenceString(
    & System.int UserPref,
    & System.int UserPrefOption,
    & System.String^ Value
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

*UserPref*User preference as defined in [swUserPreferenceStringValue_e](#)*UserPrefOption*User-preference option for customizing annotation and dimension drafting standards and as defined in [swUserPreferenceOption_e](#); if not needed, then specify swDetailingNoOptionSpecified*Value*

Value of the user preference specified in UserPreference

Return Value

True if user preference is set, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetUserPreferenceString](#).

Example

[Get and Set User Preferences \(VBA\)](#)[Get and Set User Preferences \(VB.NET\)](#)[Get and Set User Preferences \(C#\)](#)

Remarks

This method is equivalent to interactively setting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::SetUserPreferenceDouble Method](#)
[IModelDocExtension::SetUserPreferenceInteger Method](#)
[IModelDocExtension::SetUserPreferenceTextFormat Method](#)
[IModelDocExtension::SetUserPreferenceToggle Method](#)
[IModelDocExtension::GetUserPreferenceDouble Method](#)
[IModelDocExtension::GetUserPreferenceDoubleValueRange Method](#)
[IModelDocExtension::GetUserPreferenceInteger Method](#)
[IModelDocExtension::GetUserPreferenceString Method](#)
[IModelDocExtension::GetUserPreferenceTextFormat Method](#)
[IModelDocExtension::GetUserPreferenceToggle Method](#)
[ISldWorks::GetUserPreferenceDoubleValue Method](#)
[ISldWorks::GetUserPreferenceIntegerValue Method](#)
[ISldWorks::GetUserPreferenceStringListValue Method](#)
[ISldWorks::GetUserPreferenceStringValue Method](#)
[ISldWorks::GetUserPreferenceToggle Method](#)
[ISldWorks::SetUserPreferenceDoubleValue Method](#)
[ISldWorks::SetUserPreferenceIntegerValue Method](#)
[ISldWorks::SetUserPreferenceStringListValue Method](#)
[ISldWorks::SetUserPreferenceStringValue Method](#)
[ISldWorks::SetUserPreferenceToggle Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

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SetUserPreferenceTextFormat Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetUserPreferenceTextFormat Method (IModelDocExtension)

Sets document default user preference values. This method is intended for setting detailing text formats.

.NET Syntax

Visual Basic (Declaration)

```
Function SetUserPreferenceTextFormat( _  
    ByVal UserPref As System.Integer, _  
    ByVal UserPrefOption As System.Integer, _  
    ByVal Value As TextFormat _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim UserPref As System.Integer  
Dim UserPrefOption As System.Integer  
Dim Value As TextFormat  
Dim value As System.Boolean  
  
value = instance.SetUserPreferenceTextFormat(UserPref, UserPrefOption, Value)
```

C#

```
System.bool SetUserPreferenceTextFormat(  
    System.int UserPref,  
    System.int UserPrefOption,  
    TextFormat Value  
)
```

C++/CLI

```
System.bool SetUserPreferenceTextFormat(  
& System.int UserPref,  
& System.int UserPrefOption,  
& TextFormat^ Value  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPref

User preference as defined by [swUserPreferenceTextFormat_e](#)

UserPrefOption

User-preference option for customizing annotation and dimension drafting standards and as defined in [swUserPreferenceOption_e](#); if not needed, then specify swDetailingNoOptionSpecified

Value

[Text format](#) to which to assign the user preference specified in UserPref

Return Value

True if the user preference is set, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetUserPreferenceTextFormat](#).

Example

[Get and Set User Preferences \(VBA\)](#)

[Get and Set User Preferences \(VB.NET\)](#)

[Get and Set User Preferences \(C#\)](#)

Remarks

This method is equivalent to interactively setting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::SetUserPreferenceDouble Method](#)

[IModelDocExtension::SetUserPreferenceInteger Method](#)

[IModelDocExtension::SetUserPreferenceString Method](#)

[IModelDocExtension::SetUserPreferenceToggle Method](#)

[IModelDocExtension::GetUserPreferenceDouble Method](#)

[IModelDocExtension::GetUserPreferenceDoubleValueRange Method](#)

[IModelDocExtension::GetUserPreferenceInteger Method](#)

[IModelDocExtension::GetUserPreferenceString Method](#)

[IModelDocExtension::GetUserPreferenceTextFormat Method](#)

[IModelDocExtension::GetUserPreferenceToggle Method](#)

[ISldWorks::GetUserPreferenceDoubleValue Method](#)

[ISldWorks::GetUserPreferenceIntegerValue Method](#)

[ISldWorks::GetUserPreferenceStringListValue Method](#)

[ISldWorks::GetUserPreferenceStringValue Method](#)

[ISldWorks::GetUserPreferenceToggle Method](#)

[ISldWorks::SetUserPreferenceDoubleValue Method](#)

[ISldWorks::SetUserPreferenceIntegerValue Method](#)

[ISldWorks::SetUserPreferenceStringListValue Method](#)

[ISldWorks::SetUserPreferenceStringValue Method](#)

[ISldWorks::SetUserPreferenceToggle Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetUserPreferenceToggle Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetUserPreferenceToggle Method (IModelDocExtension)

Sets document default user preference values. This method is intended for user preferences that can be toggled.

.NET Syntax

Visual Basic (Declaration)

```
Function SetUserPreferenceToggle( _  
    ByVal UserPref As System.Integer, _  
    ByVal UserPrefOption As System.Integer, _  
    ByVal Value As System.Boolean _  
) As System.Boolean
```

```

Dim instance As IModelDocExtension
Dim UserPref As System.Integer
Dim UserPrefOption As System.Integer
Dim Value As System.Boolean
Dim value As System.Boolean

value = instance.SetUserPreferenceToggle(UserPref, UserPrefOption, Value)

```

C#

```

System.bool SetUserPreferenceToggle(
    System.int UserPref,
    System.int UserPrefOption,
    System.bool Value
)

```

C++/CLI

```

System.bool SetUserPreferenceToggle(
    & System.int UserPref,
    & System.int UserPrefOption,
    & System.bool Value
)

```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UserPref

User preference as defined in [swUserPreferenceToggle_e](#)

UserPrefOption

User-preference option for customizing annotation and dimension drafting standards and as defined in [swUserPreferenceOption_e](#); if not needed, then specify swDetailingNoOptionSpecified

Value

True to toggle UserPref on, false to toggle UserPref off

Return Value

True if UserPref is toggled on, false if UserPref is toggled off

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetUserPreferenceToggle](#).

Example

[Get and Set User Preferences \(VBA\)](#)
[Get and Set User Preferences \(VB.NET\)](#)
[Get and Set User Preferences \(C#\)](#)

Remarks

This method is equivalent to interactively setting document properties in the SOLIDWORKS Design software. See [System Options and Document Properties](#) for details.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::SetUserPreferenceDouble Method](#)
[IModelDocExtension::SetUserPreferenceInteger Method](#)
[IModelDocExtension::SetUserPreferenceString Method](#)
[IModelDocExtension::SetUserPreferenceTextFormat Method](#)
[IModelDocExtension::GetUserPreferenceDouble Method](#)
[IModelDocExtension::GetUserPreferenceDoubleValueRange Method](#)
[IModelDocExtension::GetUserPreferenceInteger Method](#)
[IModelDocExtension::GetUserPreferenceString Method](#)
[IModelDocExtension::GetUserPreferenceTextFormat Method](#)
[IModelDocExtension::GetUserPreferenceToggle Method](#)
[ISldWorks::GetUserPreferenceDoubleValue Method](#)
[ISldWorks::GetUserPreferenceIntegerValue Method](#)
[ISldWorks::GetUserPreferenceStringListValue Method](#)
[ISldWorks::GetUserPreferenceStringValue Method](#)

[ISldWorks::GetUserPreferenceToggle Method](#)
[ISldWorks::SetUserPreferenceDoubleValue Method](#)
[ISldWorks::SetUserPreferenceIntegerValue Method](#)
[ISldWorks::SetUserPreferenceStringListValue Method](#)
[ISldWorks::SetUserPreferenceStringValue Method](#)
[ISldWorks::SetUserPreferenceToggle Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SetVisibleBox Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SetVisibleBox Method (IModelDocExtension)

Sets the visible bounding box for Zoom to Fit for a part or an assembly.

.NET Syntax

Visual Basic (Declaration)

```
Sub SetVisibleBox( _  
    ByVal UpperLeft As MathPoint, _  
    ByVal LowerRight As MathPoint _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim UpperLeft As MathPoint  
Dim LowerRight As MathPoint  
  
instance.SetVisibleBox(UpperLeft, LowerRight)
```

C#

```
void SetVisibleBox(  
    MathPoint UpperLeft,  
    MathPoint LowerRight  
)
```

C++/CLI

```
void SetVisibleBox(  
& MathPoint^ UpperLeft,  
& MathPoint^ LowerRight  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UpperLeft

Upper-left [point](#) of the visible bounding box

LowerRight

Lower-right [point](#) of the visible bounding box

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SetVisibleBox](#).

Example

[Set Visible Bounding Box for Zoom to Fit \(C#\)](#)
[Set Visible Bounding Box for Zoom to Fit \(C#\)](#)
[Set Visible Bounding Box for Zoom to Fit \(VBA\)](#)

Remarks

Use:

- [IModelDocExtension::GetVisibleBox](#) to get the visible bounding box for a Zoom to Fit operation.

- [IModelDocExtension::RemoveVisibleBox](#) to remove the visible bounding box for a Zoom to Fit operation and to return the size of the bounding box to the size calculated by SOLIDWORKS.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ShowMessageBar Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ShowMessageBar Method (IModelDocExtension)

Shows the specified message bar in this document's window.

.NET Syntax

Visual Basic (Declaration)

```
Function ShowMessageBar( _  
    ByVal Definition As System.Object, _  
    ByVal Handler As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Definition As System.Object  
Dim Handler As System.Object  
Dim value As System.Integer  
  
value = instance.ShowMessageBar(Definition, Handler)
```

C#

```
System.int ShowMessageBar(  
    System.object Definition,  
    System.object Handler  
)
```

C++/CLI

```
System.int ShowMessageBar(  
& System.Object^ Definition,  
& System.Object^ Handler  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Definition
[IMessageBarDefinition](#)
Handler
[IMessageBarHandler](#)

Return Value

Result code as defined by [swShowMessageBarResult](#) e

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ShowMessageBar](#).

Example

See the [IMessageBarHandler](#) examples.

Remarks

This method is called by an add-in to display the specified message bar in this document's window.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

ShowModelBreakView Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ShowModelBreakView Method (IModelDocExtension)

Gets whether to show or hide the specified Model Break View in the current configuration of the active model.

.NET Syntax

Visual Basic (Declaration)

```
Function ShowModelBreakView( _  
    ByVal ShowView As System.Boolean, _  
    ByVal ViewName As System.String _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim ShowView As System.Boolean  
Dim ViewName As System.String  
Dim value As System.Boolean  
  
value = instance.ShowModelBreakView(ShowView, ViewName)
```

C#

```
System.bool ShowModelBreakView(  
    System.bool ShowView,  
    System.string ViewName  
)
```

C++/CLI

```
System.bool ShowModelBreakView(  
& System.bool ShowView,  
& System.String^ ViewName  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ShowView
True to show the Model Break View, false to hide it
ViewName
Name of Model Break View to show or hide

Return Value

True if the method executed, false if it did not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension.ShowModelBreakView](#).

Example

[Get Names and Show Model Break Views \(C#\)](#)
[Get Names and Show Model Break Views \(VB.NET\)](#)
[Get Names and Show Model Break Views \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetModelBreakViewNames Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ShowPartRebuildIndicators Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ShowPartRebuildIndicators Property (IModelDocExtension)

Gets or sets whether to display rebuild indicators on parts that have out-of-date frozen features.

.NET Syntax

Visual Basic (Declaration)

Property ShowPartRebuildIndicators As System.Boolean

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Boolean  
  
instance.ShowPartRebuildIndicators = value  
value = instance.ShowPartRebuildIndicators
```

C#

```
System.bool ShowPartRebuildIndicators {get; set;}
```

C++/CLI

```
property System.bool ShowPartRebuildIndicators {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True to display rebuild indicators, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ShowPartRebuildIndicators](#).

Example

[Move Freeze Bar \(VBA\)](#)
[Move Freeze Bar \(VB.NET\)](#)
[Move Freeze Bar \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ShowSmartMessage Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ShowSmartMessage Method (IModelDocExtension)

Displays a SOLIDWORKS-style message as a ToolTip in the graphics area and on the status bar.

.NET Syntax

Visual Basic (Declaration)

```
Sub ShowSmartMessage( _  
    ByVal Name As System.String, _  
    ByVal TimeInMillSec As System.Integer, _  
    ByVal ShowInStatusBar As System.Boolean, _  
    ByVal RemoveDefaultTip As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Name As System.String  
Dim TimeInMillSec As System.Integer  
Dim ShowInStatusBar As System.Boolean  
Dim RemoveDefaultTip As System.Boolean  
  
instance.ShowSmartMessage(Name, TimeInMillSec, ShowInStatusBar, RemoveDefaultTip)
```

C#

```
void ShowSmartMessage(  
    System.string Name,  
    System.int TimeInMillSec,  
    System.bool ShowInStatusBar,  
    System.bool RemoveDefaultTip  
)
```

C++/CLI

```
void ShowSmartMessage(  
& System.String^ Name,  
& System.int TimeInMillSec,  
& System.bool ShowInStatusBar,  
& System.bool RemoveDefaultTip  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Name
Message to display in the ToolTip
TimeInMillSec
Time, in milliseconds, to display the message
ShowInStatusBar
True to show the message on the SOLIDWORKS status bar, false to not
RemoveDefaultTip
True to replace the default SOLIDWORKS ToolTip with this message for TimeInMillSec, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ShowSmartMessage](#).

Example

Visual Basic for Applications (VBA):

Option Explicit

```
Dim swApp As SldWorks.SldWorks
Dim swModelDocExt As SldWorks.ModelDocExtension
Dim swModel As SldWorks.ModelDoc2
Dim swSelMgr As SldWorks.SelectionMgr

Sub main()

Set swApp = Application.SldWorks
Set swModel = swApp.ActiveDoc
swModel.ClearSelection2 True
Set swSelMgr = swModel.SelectionManager
Set swModelDocExt = swModel.Extension
While 1

' Loops until you select an entity in the graphics area
While swSelMgr.GetSelectedObjectCount = 0
DoEvents
Wend

swModelDocExt.ShowSmartMessage "This is the message.", 500, True, True
DoEvents
Wend

End Sub
```

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ShowUserNotification Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ShowUserNotification Method (IModelDocExtension)

Displays the specified user notification for this document.

.NET Syntax

Visual Basic (Declaration)

```
Function ShowUserNotification(  
    ByVal Definition As System.Object, _  
    ByVal Handler As System.Object _  
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Definition As System.Object  
Dim Handler As System.Object  
Dim value As System.Integer  
  
value = instance.ShowUserNotification(Definition, Handler)
```

C#

```
System.int ShowUserNotification(  
    System.object Definition,  
    System.object Handler  
)
```

C++/CLI

```
System.int ShowUserNotification(  
& System.Object^ Definition,  
& System.Object^ Handler  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Definition

[IUserNotificationDefinition](#)

Handler

[IUserNotificationHandler](#)

Return Value

Result code as defined by [swShowNotificationResult_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ShowUserNotification](#).

Example

See the [IUserNotificationHandler](#) examples.

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

SketchBoxSelect Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SketchBoxSelect Method (IModelDocExtension)

Box selects all of the entities in a sketch within the specified coordinates of the selection box.

.NET Syntax

Visual Basic (Declaration)

```
Function SketchBoxSelect( _  
    ByVal FirstPtX As System.Double, _  
    ByVal FirstPtY As System.Double, _  
    ByVal FirstPtZ As System.Double, _  
    ByVal SecondPtX As System.Double, _  
    ByVal SecondPtY As System.Double, _  
    ByVal SecondPtZ As System.Double _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim FirstPtX As System.Double  
Dim FirstPtY As System.Double  
Dim FirstPtZ As System.Double
```

```
Dim SecondPtX As System.Double
Dim SecondPtY As System.Double
Dim SecondPtZ As System.Double
Dim value As System.Boolean
```

```
value = instance.SketchBoxSelect(FirstPtX, FirstPtY, FirstPtZ, SecondPtX, SecondPtY, SecondPtZ)
```

C#

```
System.bool SketchBoxSelect(
    System.double FirstPtX,
    System.double FirstPtY,
    System.double FirstPtZ,
    System.double SecondPtX,
    System.double SecondPtY,
    System.double SecondPtZ
)
```

C++/CLI

```
System.bool SketchBoxSelect(
    & System.double FirstPtX,
    & System.double FirstPtY,
    & System.double FirstPtZ,
    & System.double SecondPtX,
    & System.double SecondPtY,
    & System.double SecondPtZ
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

FirstPtX
x coordinate of the first corner of the box
FirstPtY
y coordinate of the first corner of the box
FirstPtZ
z coordinate of the first corner of the box
SecondPtX
x coordinate of the opposite diagonal corner of the box
SecondPtY
y coordinate of the opposite diagonal corner of the box
SecondPtZ
z coordinate of the opposite diagonal corner of the box

Return Value

True if the sketch entities lying in the specified coordinates are box-selected, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SketchBoxSelect](#).

Example

[Box Select a Sketch \(VB.NET\)](#)
[Box Select a Sketch \(VBA\)](#)
[Box Select a Sketch \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0



Creates a Euclidean sketch offset from selected edges of a face or surface.

.NET Syntax

Visual Basic (Declaration)

```
Function SketchOffsetOnSurface( _  
    ByVal Offset As System.Double, _  
    ByVal Reverse As System.Boolean, _  
    ByVal UseFront As System.Boolean, _  
    ByVal MakeConstruct As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Offset As System.Double  
Dim Reverse As System.Boolean  
Dim UseFront As System.Boolean  
Dim MakeConstruct As System.Boolean  
Dim value As System.Boolean  
  
value = instance.SketchOffsetOnSurface(Offset, Reverse, UseFront, MakeConstruct)
```

C#

```
System.bool SketchOffsetOnSurface(  
    System.double Offset,  
    System.bool Reverse,  
    System.bool UseFront,  
    System.bool MakeConstruct  
)
```

C++/CLI

```
System.bool SketchOffsetOnSurface(  
& System.double Offset,  
& System.bool Reverse,  
& System.bool UseFront,  
& System.bool MakeConstruct  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Offset
Offset distance
Reverse
True to reverse the 3D sketch, false to not
UseFront
Although not currently used, must be set to true
MakeConstruct
True to make the 3D sketch construction geometry after offsetting, false to not

Return Value

True if the 3D sketch is created, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SketchOffsetOnSurface](#).

Example

[Offset Edges to Create 3D Sketch \(C#\)](#)
[Offset Edges to Create 3D Sketch \(VB.NET\)](#)
[Offset Edges to Create 3D Sketch \(VBA\)](#)

Remarks

To create a geodesic sketch offset along the curvature of a surface, use [IModelDocExtension::GeodesicSketchOffset](#).
Before calling this method, call [IModelDocExtension::SelectById2](#) for each edge to offset and set Append to true and Mark to 1.

After calling this method, call [ISketchManager::InsertSketch](#) to rebuild the model with the 3D sketch.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[ISketchManager::SketchOffset2 Method \(\)](#)
[IModelDoc2::SketchOffsetEdges Method \(\)](#)
[IModelDoc2::SketchOffsetEntities2 Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

StartFormatPainter Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : StartFormatPainter Method (IModelDocExtension)

Starts the Format Painter.

.NET Syntax

Visual Basic (Declaration)

```
Sub StartFormatPainter()
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
instance.StartFormatPainter()
```

C#

```
void StartFormatPainter()
```

C++/CLI

```
void StartFormatPainter();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::StartFormatPainter](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::ApplyFormatPainterToAll Method](#)
[IModelDocExtension::StopFormatPainter Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

StartRecordingUndoObject Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : StartRecordingUndoObject Method (IModelDocExtension)

Starts recording the SOLIDWORKS Undo object.

.NET Syntax

```
Sub StartRecordingUndoObject()
```



```
Dim instance As IModelDocExtension
```

```
instance.StartRecordingUndoObject()
```

```
void StartRecordingUndoObject()
```

```
void StartRecordingUndoObject();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::StartRecordingUndoObject](#).

Example

[Create Hidden Undo Object \(VBA\)](#)
[Create Hidden Undo Object \(VB.NET\)](#)
[Create Hidden Undo Object \(C#\)](#)

Remarks

To add an object with the name UndoObjectName to the SOLIDWORKS Undo list, place the SOLIDWORKS API calls between IModelDocExtension::StartRecordingUndoObject and [IModelDocExtension::FinishRecordingUndoObject2](#):

For example, assume that your application creates a complex gear that requires many SOLIDWORKS API calls and you would like your SOLIDWORKS Design user to be able to undo that gear from the user interface at a later time. Your application can create an Undo object by calling:

1. IModelDocExtension::StartRecordingUndoObject
2. SOLIDWORKS API calls that create the gear
3. IModelDocExtension::FinishRecordingUndoObject2 with MakeHidden = False

Now assume that your application performs many temporary operations that you do not want your users to see in the SOLIDWORKS Undo list, because they are not essential to the modeling and design process. For this case, your application can place the non-essential SOLIDWORKS API calls between IModelDocExtension::StartRecordingUndoObject and IModelDocExtension::FinishRecordingUndoObject2 with MakeHidden = True. The Undo object is still added to the SOLIDWORKS Undo list, but the user does not see it in the list.

WARNING: If your application hides an Undo object in the Undo list by setting MakeHidden = True and the Undo object is the first item in the SOLIDWORKS Undo list, then SOLIDWORKS discards the Undo object from the Undo list. If this happens, the SOLIDWORKS API calls in this Undo object cannot be undone.

Similarly, if the last item to be redone is a hidden API Undo object, then SOLIDWORKS discards it, and the SOLIDWORKS API calls cannot be redone.

Only SOLIDWORKS operations that support Undo are undone. Both SOLIDWORKS API and non-API operations are undone.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::ClearUndoList Method](#)
[IModelDoc2::EditRedo2 Method](#)

Availability

SOLIDWORKS 2007 FCS, Revision Number 15.0



Stops the Format Painter.

.NET Syntax

Visual Basic (Declaration)

```
Sub StopFormatPainter()
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
instance.StopFormatPainter()
```

C#

```
void StopFormatPainter()
```

C++/CLI

```
void StopFormatPainter();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::StopFormatPainter](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::StartFormatPainter Method](#)
[IModelDocExtension::ApplyFormatPainterToAll Method](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Stretch Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : Stretch Method (IModelDocExtension)

Stretch the selected entities.

.NET Syntax

Visual Basic (Declaration)

```
Sub Stretch( _  
    ByVal KeepRelations As System.Boolean, _  
    ByVal BaseX As System.Double, _  
    ByVal BaseY As System.Double, _  
    ByVal DestX As System.Double, _  
    ByVal DestY As System.Double _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim KeepRelations As System.Boolean  
Dim BaseX As System.Double  
Dim BaseY As System.Double  
Dim DestX As System.Double  
Dim DestY As System.Double
```

```
instance.Stretch(KeepRelations, BaseX, BaseY, DestX, DestY)
```

C#

```
void Stretch(  
    System.bool KeepRelations,  
    System.double BaseX,  
    System.double BaseY,  
    System.double DestX,  
    System.double DestY  
)
```

C++/CLI

```
void Stretch(  
& System.bool KeepRelations,  
& System.double BaseX,  
& System.double BaseY,  
& System.double DestX,  
& System.double DestY  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

KeepRelations

True to keep the sketch relations intact, false to sever them

BaseX

x coordinate of the base point

BaseY

y coordinate of the base point

DestX

x coordinate of the destination stretch

DestY

y coordinate of the destination of the stretch

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::Stretch](#).

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

SunLightInformation Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : SunLightInformation Property (IModelDocExtension)

Gets the specified sunlight information.

.NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property SunLightInformation( _  
    ByVal Type As System.Integer _  
) As System.Double
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim Type As System.Integer  
Dim value As System.Double
```



```
value = instance.SunLightInformation(Type)
```

C#

```
System.double SunLightInformation(  
    System.int Type  
) {get;}
```

C++/CLI

```
property System.double SunLightInformation {  
    System.double get(System.int Type);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Type
Sunlight information as defined by [swSunlightInfoType_e](#)

Property Value

Sunlight information

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::SunLightInformation](#).

Example

[Get and Set Sunlight Source Property Values \(VBA\)](#)
[Get and Set Sunlight Source Property Values \(VB.NET\)](#)
[Get and Set Sunlight Source Property Values \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::UpdateSunLight Method \(\)](#)
[IModelDocExtension::GetSunLightAdvancedPropertyValues Method \(\)](#)
[IModelDocExtension::SetSunLightAdvancedPropertyValues Method \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

ToolboxPartType Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ToolboxPartType Property (IModelDocExtension)

Gets and sets whether this part is a SOLIDWORKS Toolbox part.

.NET Syntax

Visual Basic (Declaration)

```
Property ToolboxPartType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Integer
```

```
instance.ToolboxPartType = value
```

```
value = instance.ToolboxPartType
```

C#

```
System.int ToolboxPartType {get; set;}
```

C++/CLI

```
property System.int ToolboxPartType {  
    System.int get();  
    void set ( & System.int value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Type of SOLIDWORKS Toolbox part type as defined in [swToolBoxPartType_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ToolBoxPartType](#).

Example

[Test for Toolbox Part \(VBA\)](#)
[Test for Toolbox Part \(VB.NET\)](#)
[Test for Toolbox Part \(C#\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

Transparency Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : Transparency Property (IModelDocExtension)

Gets and sets the transparency states for the specified display state setting.

.NET Syntax

Visual Basic (Declaration)

```
Property Transparency( _  
    ByVal Setting As DisplayStateSetting _  
) As System.Object
```

Visual Basic (Usage)

```
Dim instance As IModelDocExtension  
Dim Setting As DisplayStateSetting  
Dim value As System.Object
```

```
instance.Transparency(Setting) = value
```

```
value = instance.Transparency(Setting)
```

C#

```
System.object Transparency(  
    DisplayStateSetting Setting  
) {get; set;}
```

C++/CLI



```
property System.Object^ Transparency {
    System.Object^ get(DisplayStateSetting^ Setting);
    void set (DisplayStateSetting^ Setting, System.Object^ value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Setting
[IDisplayStateSetting](#)

Property Value

One-dimensional array of transparency states as defined in [swTransparencyState_e](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::Transparency](#).

Example

[Get Display State Settings \(C#\)](#)
[Get Display State Settings \(VB.NET\)](#)
[Get Display State Settings \(VBA\)](#)

Remarks

Before calling this method:

1. Call [IModelDocExtension::GetDisplayStateSetting](#) to obtain an [IDisplayStateSetting](#) object.
2. Call [IDisplayStateSetting::Names](#) or [IDisplayStateSetting::SetNames](#) to specify the display states.
3. Call [IDisplayStateSetting::Entities](#) or [IDisplayStateSetting::SetEntities](#) to specify the components.
4. Call [IDisplayStateSetting::Option](#) to specify the display state scope of the setting.
5. Specify the Setting parameter using the IDisplayStateSetting object.

Each transparency state in the array returned by this method maps to a component in a display state. The size of the returned array is ([IDisplayStateSetting::GetEntityCount](#) X [IDisplayStateSetting::GetNameCount](#)).

The returned array stores transparency states (TS) for N components in M display states as follows:

Component1 DisplayState1 TS, Component1 DisplayState2 TS, ..., Component1 DisplayStateM TS,
 Component2 DisplayState1 TS, Component2 DisplayState2 TS, ..., Component2 DisplayStateM TS,
 ...,
 ComponentN DisplayState1 TS, ComponentN DisplayState2 TS, ..., ComponentN DisplayStateM TS

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::DisplayMode Property](#)
[IModelDocExtension::DisplayStateSpecMaterialPropertyValues Property](#)
[IModelDocExtension::Visibility Property](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UpdateExternalFileReferences Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : UpdateExternalFileReferences Method (IModelDocExtension)

Updates the external files references on this model.

.NET Syntax

Visual Basic (Declaration)

```
Sub UpdateExternalFileReferences( _
    ByVal ConfigOption As System.Integer, _
    ByVal ConfigName As System.String, _
```

```
ByVal UpdateStatus As System.Integer _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim ConfigOption As System.Integer  
Dim ConfigName As System.String  
Dim UpdateStatus As System.Integer
```

```
instance.UpdateExternalFileReferences(ConfigOption, ConfigName, UpdateStatus)
```

C#

```
void UpdateExternalFileReferences(  
    System.int ConfigOption,  
    System.string ConfigName,  
    System.int UpdateStatus  
)
```

C++/CLI

```
void UpdateExternalFileReferences(  
& System.int ConfigOption,  
& System.String^ ConfigName,  
& System.int UpdateStatus  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ConfigOption

Configuration options as defined by [swExternalFileReferencesConfig_e](#)

ConfigName

Name of configuration; required when configOption set to swExternalFileReferencesNamedConfig

UpdateStatus

Update status option as defined by [swExternalFileReferencesUpdate_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::UpdateExternalFileReferences](#).

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::ListExternalFileReferences Method](#)

[IModelDocExtension::ListExternalFileReferencesCount Method](#)

[IModelDocExtension::ListExternalFileReferences Method](#)

[IModelDoc2::BreakAllExternalReferences Method](#)

[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)

[IModelDoc2::ListAuxiliaryExternalFileReferences Method](#)

[IModelDoc2::ListAuxiliaryExternalFileReferencesCount Method](#)

[IModelDoc2::LockAllExternalReferences Method](#)

Availability

SOLIDWORKS 2004 FCS, Revision Number 12.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

UpdateFrozenFeatures Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : UpdateFrozenFeatures Method (IModelDocExtension)

Updates frozen features of the model.

.NET Syntax

Visual Basic (Declaration)

```
Function UpdateFrozenFeatures( _  
    ByVal UpdateAllConfigs As System.Boolean _
```

```
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim UpdateAllConfigs As System.Boolean  
Dim value As System.Integer  
  
value = instance.UpdateFrozenFeatures(UpdateAllConfigs)
```

C#

```
System.int UpdateFrozenFeatures(  
    System.bool UpdateAllConfigs  
)
```

C++/CLI

```
System.int UpdateFrozenFeatures(  
&    System.bool UpdateAllConfigs  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UpdateAllConfigs
True to update all configurations, false to not

Return Value

Number of frozen features updated

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::UpdateFrozenFeatures](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IFeature::HasFrozenUpdatePending Method](#)
[IFeature::IsFrozen Method](#)
[IFeature::MoveFreezeBarTo Method](#)
[IFeatureManager::EditFreeze Method](#)
[IFeatureManager::GetFreezeLocation Method](#)
[IModelDocExtension::NeedsRebuild2 Property](#)
[IFeature::IsHiddenLock Method](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

UpdateRenderMaterialsInSceneGraph Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : UpdateRenderMaterialsInSceneGraph Method (IModelDocExtension)

Sets whether to update the appearances in the graphics area in the model.

.NET Syntax

Visual Basic (Declaration)

```
Sub UpdateRenderMaterialsInSceneGraph( _  
    ByVal AddToSG As System.Boolean _  
)
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim AddToSG As System.Boolean
```

```
instance.UpdateRenderMaterialsInSceneGraph(AddToSG)
```

C#

```
void UpdateRenderMaterialsInSceneGraph(  
    System.bool AddToSG  
)
```

C++/CLI

```
void UpdateRenderMaterialsInSceneGraph(  
& System.bool AddToSG  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

AddToSG

True to update the appearances in the graphics area, false to not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::UpdateRenderMaterialsInSceneGraph](#).

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::AddDefaultRenderMaterial Method](#)
[IModelDocExtension::AddRenderMaterial Method](#)
[IModelDocExtension::CreateRenderMaterial Method](#)
[IModelDocExtension::DeleteRenderMaterial Method](#)
[IModelDocExtension::GetMaterial Method](#)
[IModelDocExtension::GetRenderMaterials Method](#)
[IModelDocExtension::GetRenderMaterialsCount Method](#)
[IModelDocExtension::IGetRenderMaterials Method](#)
[IRenderMaterial Interface](#)

Availability

SOLIDWORKS 2008 FCS, Revision Number 16.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

UpdateStandardViews Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : UpdateStandardViews Method (IModelDocExtension)

Changes the specified standard view to the current model view.

.NET Syntax

Visual Basic (Declaration)

```
Function UpdateStandardViews( _  
    ByVal ViewName As System.String, _  
    ByVal ViewId As System.Integer _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim ViewName As System.String  
Dim ViewId As System.Integer  
Dim value As System.Boolean  
  
value = instance.UpdateStandardViews(ViewName, ViewId)
```

C#

```
System.bool UpdateStandardViews(  
    System.string ViewName,  
    System.int ViewId  
)
```

C++/CLI

```
System.bool UpdateStandardViews(  
& System.String^ ViewName,  
& System.int ViewId  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

ViewName

Name of the standard model view to change; empty string to use ViewId (see **Remarks**)

ViewId

View ID as defined in [swStandardViews_e](#); -1 to use ViewName (see **Remarks**)

Return Value

True if the standard model view update is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::UpdateStandardViews](#).

Remarks

This method works with parts and assemblies only.

If you set both ViewName and ViewId, then ViewId takes precedence if the two arguments do not resolve to the same standard model view.

This method does not support the following standard model views:

- [swStandardViews_e.swDimetricView](#)
- [swStandardViews_e.swIsometricView](#)
- [swStandardViews_e.swTrimetricView](#)

See Also

[IModelDocExtension Interface](#)

[IModelDocExtension Members](#)

[IModelDocExtension::ResetStandardViews Method \(\)](#)

[IModelDoc2::ShowNamedView2 Method \(\)](#)

Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UpdateSunLight Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : UpdateSunLight Method (IModelDocExtension)

Updates sunlight position, color, and background image.

.NET Syntax

Visual Basic (Declaration)

```
Function UpdateSunLight() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Boolean
```

```
value = instance.UpdateSunLight()
```

C#

```
System.bool UpdateSunLight()
```

C++/CLI

```
System.bool UpdateSunLight();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if the update is successful, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::UpdateSunLight](#).

Remarks

Before calling this method, call [IModelDocExtension::SetSunLightAdvancedPropertyValues](#) to set sunlight advanced properties.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::GetSunLightAdvancedPropertyValues Method \(\)](#)
[IModelDocExtension::SunLightInformation Property \(\)](#)

Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UpgradeLegacyCThreads Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : UpgradeLegacyCThreads Method (IModelDocExtension)

Upgrades cosmetic thread features in this legacy model to the latest cosmetic thread architecture.

.NET Syntax

Visual Basic (Declaration)

```
Function UpgradeLegacyCThreads() As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Boolean  
  
value = instance.UpgradeLegacyCThreads()
```

C#

```
System.bool UpgradeLegacyCThreads()
```

C++/CLI

```
System.bool UpgradeLegacyCThreads();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Return Value

True if cosmetic threads upgraded successfully in this legacy model, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::UpgradeLegacyCThreads](#).

Remarks

This method:

- is valid only if the system option allowing the upgrade of legacy files containing cosmetic threads is selected.
- corresponds to the user-interface command, *feature_manager_design_tree_top_node* **RMB > Upgrade cosmetic thread features**.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::HasLegacyCThreads Method \(\)](#)

Availability

SOLIDWORKS 2019 FCS, Revision Number 27.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

UpgradeLegacyCustomProperties Method (IModelDocExtension)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : UpgradeLegacyCustomProperties Method (IModelDocExtension)

Upgrades custom properties in this legacy (created prior to SOLIDWORKS 2018) model to the latest custom properties architecture.

.NET Syntax

Visual Basic (Declaration)

```
Function UpgradeLegacyCustomProperties(  
    ByVal UpgradeAllAssemComps As System.Boolean _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim UpgradeAllAssemComps As System.Boolean  
Dim value As System.Boolean  
  
value = instance.UpgradeLegacyCustomProperties(UpgradeAllAssemComps)
```

C#

```
System.bool UpgradeLegacyCustomProperties(  
    System.bool UpgradeAllAssemComps  
)
```

C++/CLI

```
System.bool UpgradeLegacyCustomProperties(  
& System.bool UpgradeAllAssemComps  
)
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

UpgradeAllAssemComps

True to upgrade custom properties of this top-level assembly and all of its components, false to upgrade custom properties of this top-level assembly only

Return Value

True if custom properties are successfully upgraded, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::UpgradeLegacyCustomProperties](#).

Remarks

This method is valid only for parts, assemblies, and drawings:

- created prior to 2018.
- opened in resolved mode.
- corresponds to the user-interface command, *feature_manager_design_tree_top_node* **RMB > Upgrade custom properties**.

This method upgrades the custom properties of:

- parent parts only, not derived parts
- drawings, but not models in drawing views
- weldment cutlists
- (optionally) all components in assemblies

After running this method, you:

- must rebuild configurations to have their custom properties updated.
- cannot return to the old custom properties architecture.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)

Availability

SOLIDWORKS 2022 FCS, Revision Number 30

SOLIDWORKS API Help

[Send comments on this topic.](#)

UsePageSetup Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : UsePageSetup Property (IModelDocExtension)

Gets or sets whether this document uses its own page setup values, SOLIDWORKS Design application page setup values, or setup values on individual drawing sheets.

.NET Syntax

Visual Basic (Declaration)

Property UsePageSetup As System.Integer

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
Dim value As System.Integer
```

```
instance.UsePageSetup = value
```

```
value = instance.UsePageSetup
```

C#

```
System.int UsePageSetup {get; set;}
```

C++/CLI

```
property System.int UsePageSetup {
    System.int get();
    void set ( & System.int value);
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Page setup to use as defined in [swPageSetupInUse_e](#)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::UsePageSetup](#).

Example

[Print Drawing \(C#\)](#)
[Print Drawing \(VB.NET\)](#)
[Print Drawing \(VBA\)](#)

Remarks

To retrieve this page setup...	Use...
Application	IModelDocExtension::AppPageSetup
Document	IModelDoc2::PageSetup or IModelDoc2::IPageSetup
Sheet	ISheet::PageSetup or ISheet::IPageSetup

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IPageSetup Interface](#)

Availability

SOLIDWORKS 2003 FCS, Revision Number 11.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ViewDisplayRealView Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ViewDisplayRealView Property (IModelDocExtension)

Gets or sets the RealView Graphics setting.

.NET Syntax

Visual Basic (Declaration)

Property ViewDisplayRealView As System.Boolean

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim value As System.Boolean  
  
instance.ViewDisplayRealView = value  
value = instance.ViewDisplayRealView
```

C#

```
System.bool ViewDisplayRealView {get; set;}
```

C++/CLI

```
property System.bool ViewDisplayRealView {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if RealView Graphics is set, false if not

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ViewDisplayRealView](#).

Example

[Get Appearance File Name \(C#\)](#)
[Get Appearance File Name \(VB.NET\)](#)
[Get Appearance File Name \(VBA\)](#)

Remarks

When you [apply a material to a part](#), use RealView Graphics to display the model with realistic-looking materials. RealView Graphics is available with supported graphics cards only. See the SOLIDWORKS Help for details.

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IMaterialVisualPropertiesData::RealView Property](#)

Availability

SOLIDWORKS 2005 FCS, Revision Number 13.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

ViewZoomToSheet Method (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : ViewZoomToSheet Method (IModelDocExtension)

Zooms a drawing sheet to its maximum size within the window.

.NET Syntax

Visual Basic (Declaration)

```
Sub ViewZoomToSheet()
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension
```

```
instance.ViewZoomToSheet()
```

C#

```
void ViewZoomToSheet()
```

C++/CLI

```
void ViewZoomToSheet();
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::ViewZoomToSheet](#).

Example

[Zoom Drawing Sheet to Maximum Size in Window \(C#\)](#)
[Zoom Drawing Sheet to Maximum Size in Window \(VB.NET\)](#)
[Zoom Drawing Sheet to Maximum Size in Window \(VBA\)](#)

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDoc2::ViewOrientationUndo Method \(\)](#)
[IModelDoc2::ViewRotate Method \(\)](#)
[IModelDoc2::ViewRotateminusx Method \(\)](#)
[IModelDoc2::ViewRotateminusy Method \(\)](#)
[IModelDoc2::ViewRotateminusz Method \(\)](#)
[IModelDoc2::ViewRotateplusx Method \(\)](#)
[IModelDoc2::ViewRotateplusy Method \(\)](#)
[IModelDoc2::ViewRotateplusz Method \(\)](#)
[IModelDoc2::ViewRotXMinusNinety Method \(\)](#)
[IModelDoc2::ViewRotXPlusNinety Method \(\)](#)
[IModelDoc2::ViewRotYMinusNinety Method \(\)](#)
[IModelDoc2::ViewRotYPlusNinety Method \(\)](#)
[IModelDoc2::ViewZoomin Method \(\)](#)
[IModelDoc2::ViewZoomout Method \(\)](#)
[IModelDoc2::ViewZoomto Method \(\)](#)

[IModelDoc2::ViewZoomTo2 Method \(\)](#)
[IModelDoc2::ViewZoomToFit2 Method \(\)](#)
[IModelDoc2::ViewZoomToSelection Method \(\)](#)

Availability

SOLIDWORKS 2017 FCS, Revision Number 25.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

Visibility Property (IModelDocExtension)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IModelDocExtension Interface](#) : Visibility Property (IModelDocExtension)

Gets and sets the visibility states for the specified display state setting.

.NET Syntax

Visual Basic (Declaration)

```
Property Visibility( _  
    ByVal Setting As DisplayStateSetting _  
) As System.Object
```

Visual Basic (Usage)



```
Dim instance As IModelDocExtension  
Dim Setting As DisplayStateSetting  
Dim value As System.Object  
  
instance.Visibility(Setting) = value  
value = instance.Visibility(Setting)
```

C#

```
System.object Visibility(  
    DisplayStateSetting Setting  
) {get; set;}
```

C++/CLI

```
property System.Object^ Visibility {  
    System.Object^ get(DisplayStateSetting^ Setting);  
    void set (DisplayStateSetting^ Setting, System.Object^ value);  
}
```

NOTE: See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Parameters

Setting
[IDisplayStateSetting](#)

Property Value

One-dimensional array of visibility states as defined in [swVisibilityState_e](#) (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [ModelDocExtension::Visibility](#).

Example

[Get Display State Settings \(VBA\)](#)
[Get Display State Settings \(VB.NET\)](#)
[Get Display State Settings \(C#\)](#)

Remarks

Before calling this method:

1. Call [IModelDocExtension::GetDisplayStateSetting](#) to obtain an [IDisplayStateSetting](#).

2. Call [IDisplayStateSetting::Names](#) or [IDisplayStateSetting::ISetNames](#) to specify the display states.
3. Call [IDisplayStateSetting::Entities](#) or [IDisplayStateSetting::ISetEntities](#) to specify the components.
4. Call [IDisplayStateSetting::Option](#) to specify the display state scope of the setting.
5. Specify the Setting parameter using the IDisplayStateSetting object.

Each visibility state in the array returned by this method maps to a component in a display state. The size of the returned array is ([IDisplayStateSetting::GetEntityCount](#) X [IDisplayStateSetting::GetNameCount](#)).

The returned array stores visibility states (VS) for N components in M display states as follows:

Component1 DisplayState1 VS, Component1 DisplayState2 VS, ..., Component1 DisplayStateM VS,
Component2 DisplayState1 VS, Component2 DisplayState2 VS, ..., Component2 DisplayStateM VS,
...,
ComponentN DisplayState1 VS, ComponentN DisplayState2 VS, ..., ComponentN DisplayStateM VS

See Also

[IModelDocExtension Interface](#)
[IModelDocExtension Members](#)
[IModelDocExtension::DisplayMode Property](#)
[IModelDocExtension::DisplayStateSpecMaterialPropertyValues Property](#)
[IModelDocExtension::Transparency Property](#)

Availability

SOLIDWORKS 2012 FCS, Revision Number 20.0